

Soham Shirke

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EDUCATION

Bhartiya Vidya Bhavan's Sardar Patel Institute of Technology <i>Bachelor of Technology in Computer Science and Engineering (Specialization: AI & ML)</i>	Andheri, Mumbai 2022 – 2026
B.K. Birla College of Arts, Science and Commerce <i>HSC (Class XII), Maharashtra State Board — 93%</i>	Kalyan, Thane 2021
Mahila Samiti English High School and Junior College <i>SSC (Class X), Maharashtra State Board — 86.20%</i>	Thakuri, Thane 2019

RELEVANT COURSEWORK

- Data Structures
- Database Management
- Blockchain Technology
- Operating Systems
- Algorithms Analysis
- Artificial Intelligence
- Machine Learning
- Computer Architecture

PROJECTS

- ML for AOP Retrieval from Geostationary Satellites** | Python, PyTorch, Satellite Data Feb 2025
- Ongoing research project focusing on aerosol property retrieval using machine learning techniques applied to geostationary satellite data.
 - Implemented the NNAeroG architecture using PyTorch to replicate baseline performance from the original paper.
 - Used JAXA's Himawari-8 satellite data as input and validated outputs using AERONET ground truth station data.
 - Currently investigating novel model architectures and enhancements to improve retrieval accuracy and identify research contributions.
- Diet Recommendation System** | Python, Django, Scikit-learn, Joblib, Selenium | [GitHub](#) Apr 2024
- Engineered the backend of a personalized diet recommendation web application using Django and ML.
 - Trained and deployed a Scikit-learn model to suggest dietary plans based on user attributes like age, weight, and health goals.
 - Automated food image retrieval using a custom script with Selenium and Chromedriver.
 - Serialized and served model predictions via REST API using Django views and Joblib.

TECHNICAL SKILLS

Programming Languages: Python, C, JavaScript, SQL
Web Technologies: HTML, CSS, React, Django
Frameworks & Libraries: NumPy, Pandas, TensorFlow, Matplotlib, Dialogflow, YOLO, BeautifulSoup
Developer Tools: VS Code, Google Colab, Google Cloud Platform, Amazon AWS, GitHub, Docker
Technologies & Platforms: Linux (Ubuntu), WordPress, Figma
Languages: English, Hindi, Marathi

HOBBIES

- Cricket
- Listening to Music
- Video games